

Public Finance

BIEM – Spring 2026

Taxes on Savings (Ch.22)



Taxes on Savings

- Taxes on capital income affect how individuals allocate consumption over time.
- Understanding these responses is essential for evaluating the efficiency and distributional consequences of tax policy in a context where many countries worry about insufficient private saving for retirement.
- This lecture follows the core structure in Gruber Ch.22 and integrates recent evidence and an Italian/European perspective.



What do we mean by taxing capital income?

Type of taxation	What is taxed	Examples
Taxes on savings	The return to saving over time	Interest income
Taxes on capital income	Financial returns from assets	Interest, dividends, capital gains
Taxes on wealth transfers and property	Transfer or ownership of assets	Inheritance, gifts, property taxes



Overview

- Individuals choose how much to consume today versus in the future.
- Savings is the mechanism linking today's consumption to tomorrow's consumption.
- Taxes on capital income reduce the after-tax return to saving and may alter this allocation.
- As in labor supply, the key forces are:
 - **Substitution effects** (saving becomes less attractive)
 - **Income effects** (lifetime resources fall)
- Theory predicts **ambiguous** total effects on saving.
- Empirical evidence has struggled to settle the debate.



Traditional Theory

- The intertemporal choice model is the main model for understanding how taxes affect savings.
 - Intertemporal choice model: A choice about how much to save is really a choice about how to allocate one's consumption over time.
 - Savings is not directly modeled but is a residual: the excess of current income over current consumption.
- The model focuses on the trade-off between consumption today and consumption tomorrow.



Intertemporal choice model (1)

Suppose Minh lives for two periods.

- Period 1 is his working life, during which she earns income Y .
- Period 2 is her retirement, during which she earns nothing.
- She chooses how much to consume while working (c^W) and while retired (c^R).
- Savings is the difference between income and first-period consumption:

$$S = Y - c^W$$

- Savings earn interest, and Minh consumes her net savings:

$$^5(1 + r)S = c^R = (1 + r)(Y - c^W)$$



Intertemporal choice model (2)

- Intertemporal budget constraint: The measure of the rate at which individuals can trade off consumption in one period for consumption in another period:

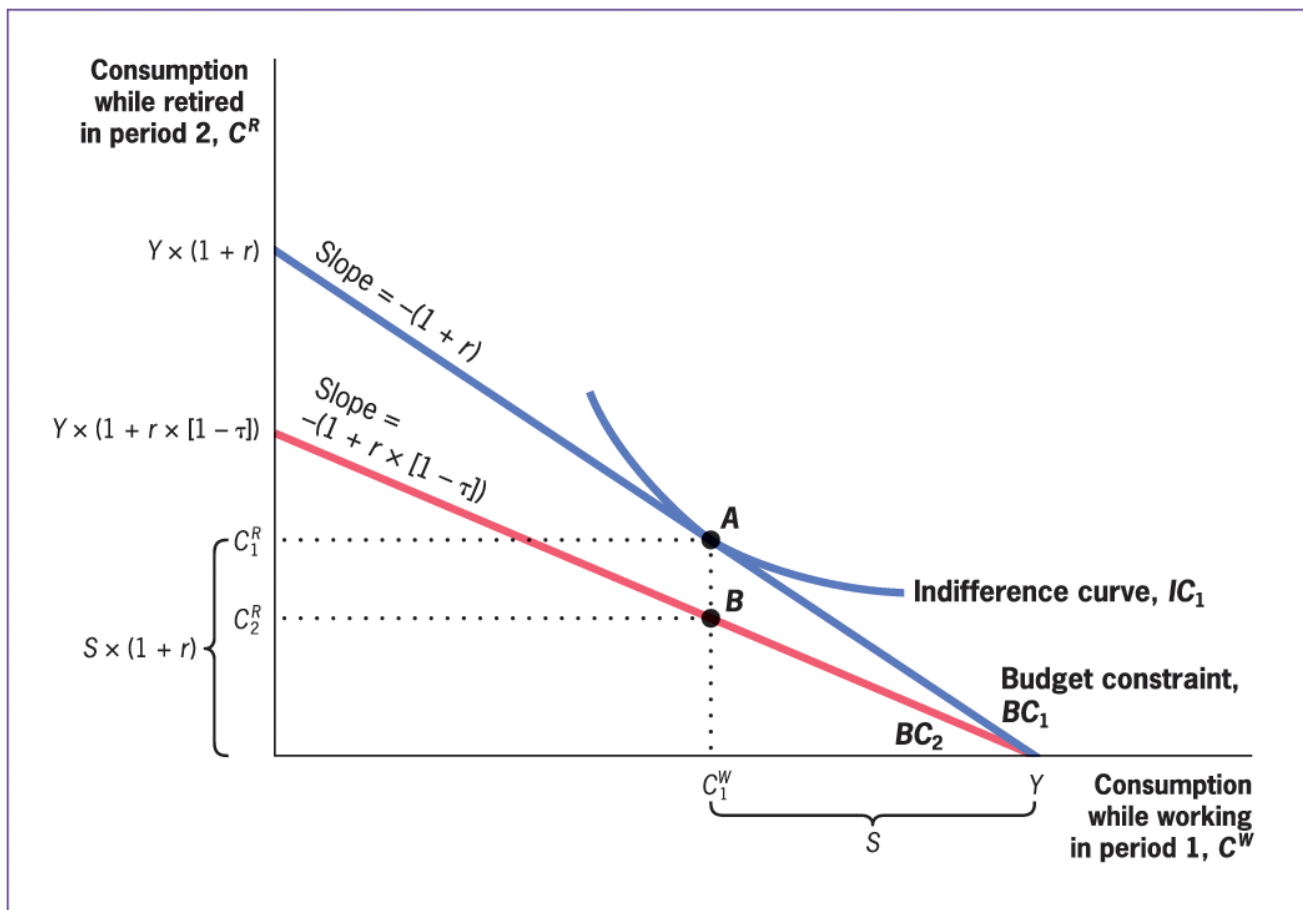
$$C^R = (1 + R)(Y - C^W)$$

- The opportunity cost of one dollar of first-period consumption is $(1 + r)$ dollars of second-period consumption.
- Taxes on savings affect behavior by changing the effective interest rate, shifting the budget constraint.



Intertemporal choice model (3)

Figure 22-1



© Worth Publishers

Before taxes are introduced, individuals lose $-(1+r)$ worth of consumption in period 2 for every dollar of consumption in period 1. Individuals save S in the first period and consume $S \times (1+r)$ in the second period.

When taxes rise, the budget constraint pivots inward, and individuals lose $-(1+r \times [1-t])$ worth of consumption in period 2 for \$1 of consumption in period 1. Savings may rise or fall depending on the relative magnitudes of the income and substitution effects.



Substitution and Income Effects of Taxes on Savings

Taxes and other price changes affect savings in two ways:

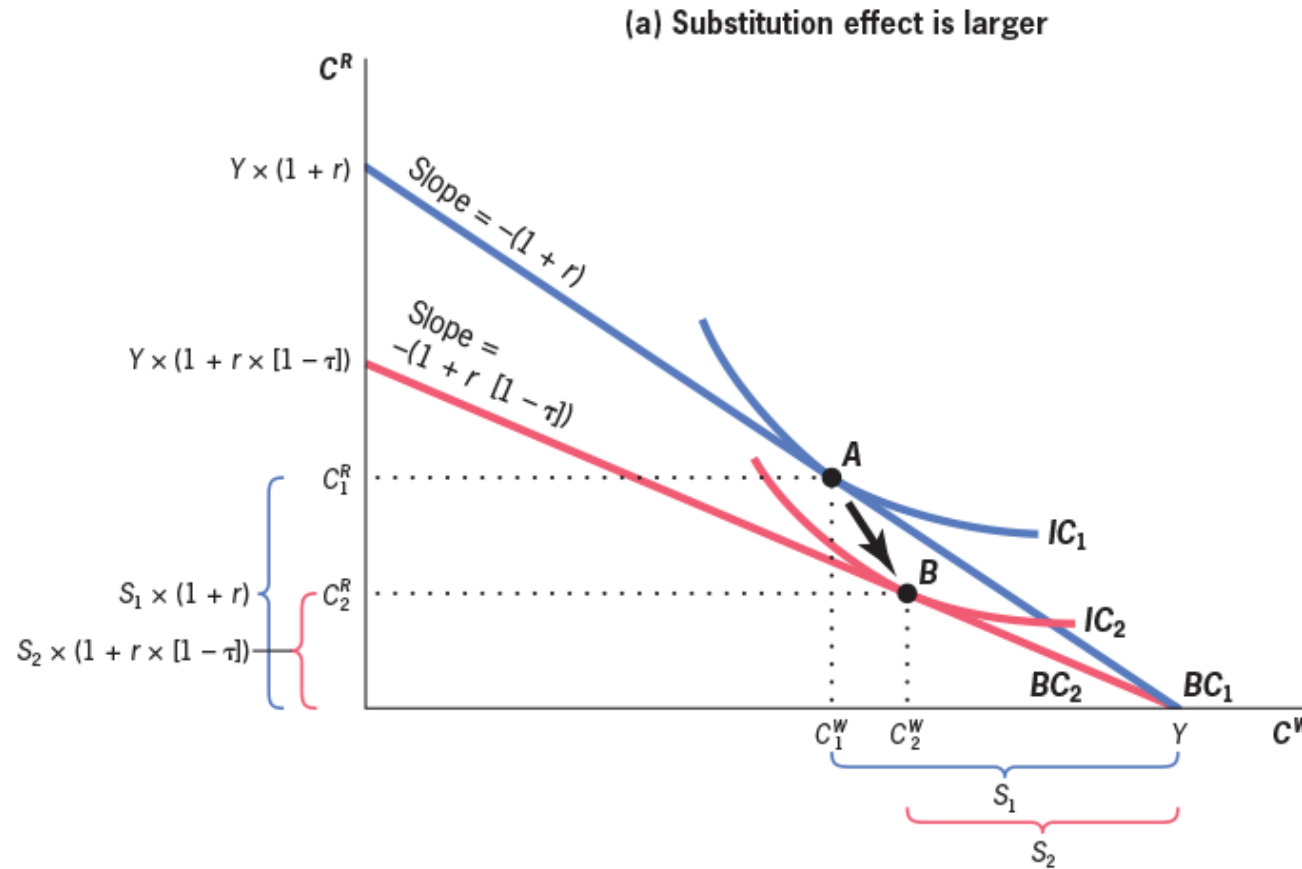
- **Substitution effect:** Lower after-tax interest rates cause first-period consumption to rise, reducing savings.
- **Income effect:** Lower after-tax interest rates reduce the lifetime value of income, reducing first-period consumption and increasing savings.

Substitution effects may seem more natural, but a “target savings” model generates complete income effects.



Effect of Taxes on Savings: Substitution Effect Is Larger

Figure 22-2a



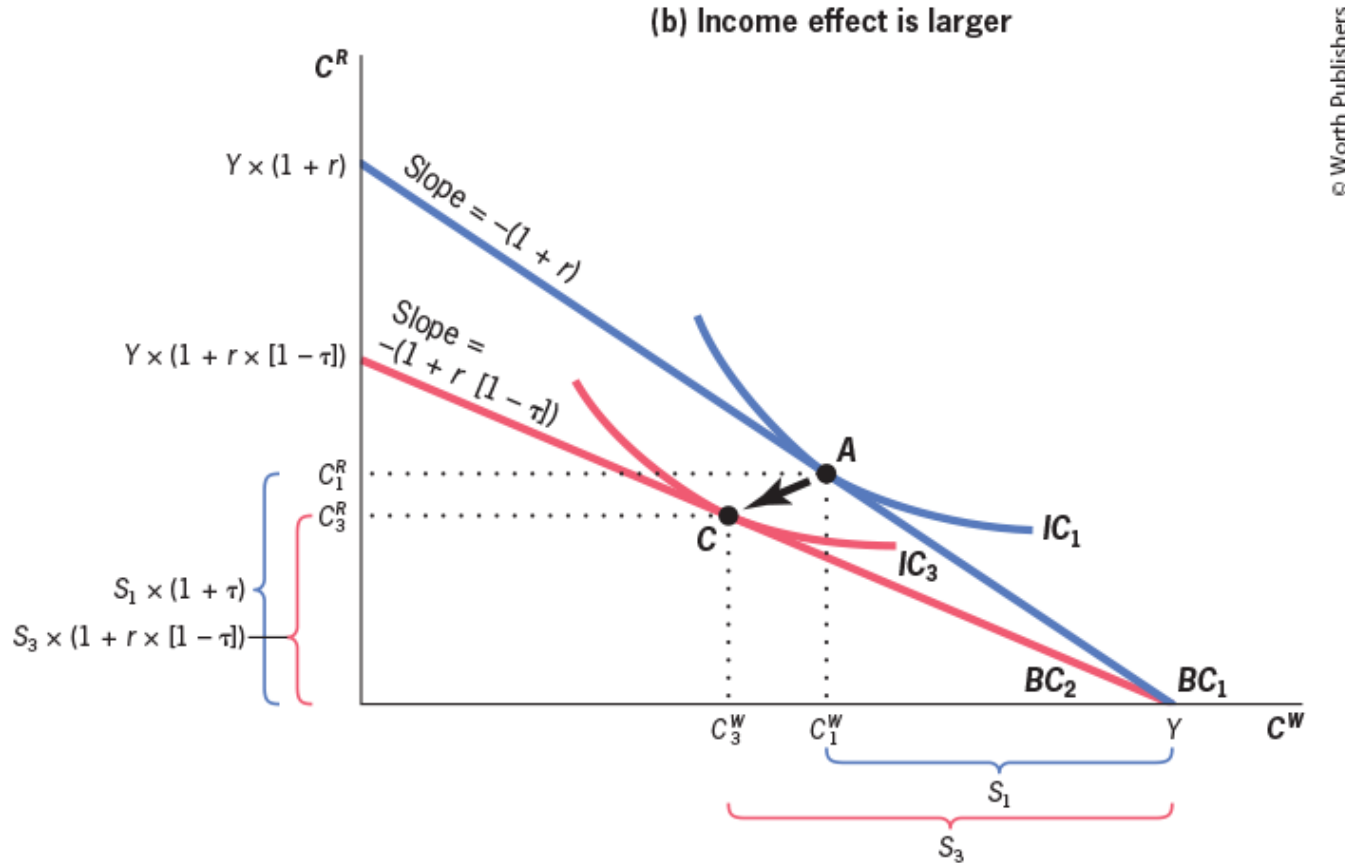
© Worth Publishers

If the substitution effect is larger than the income effect, individuals will move from point A to point B, consuming more in the first period (CW_2) and thus saving less (S_2). As a result, their consumption in period 2 (CR_2) falls by a lot.



Effect of Taxes on Savings: Income Effect Is Larger

Figure 22-2b



If the income effect is larger, individuals will move from point A to point C, consuming less in the first period (CW_3) and thus saving more (S_3). Their consumption in period 2 (CR_3) still falls but not by as much.



The intuition behind the substitution and income effect

A lower after-tax interest rate reduces the reward from postponing consumption.

Substitution effect:

- Future consumption becomes relatively cheaper.
- Individuals substitute **toward more consumption today**.
- Therefore **saving falls**.

This is the intuitive channel: taxing capital income makes saving less attractive.

A lower after-tax return reduces lifetime resources.

Income effect:

- Individuals feel poorer overall.
- They reduce consumption in all periods.
- Lower C_1 *raises saving*.

This effect pushes in the **opposite direction** of the substitution effect.



Ambiguous Impact of Taxes on Savings

Because the substitution and income effects move in opposite directions:

The net effect of a tax on capital income is theoretically ambiguous.

Depending on preferences:

- If substitution effect $>$ income effect $\rightarrow$ **savings fall.**
- If income effect $>$ substitution effect $\rightarrow$ **savings rise.**



Difficulties in Measuring Elasticities

Empirically estimating how savings responds to after-tax interest rates is challenging:

- Identifying the "relevant" interest rate is difficult.
- Interest rates tend to vary at the same time for everyone, making clean treatment-control comparisons rare.
- Many types of savings instruments have different returns and tax treatments.
- Long horizons complicate causal inference.

As a result, economists disagree on whether taxes reduce, increase, or do not affect total private saving.



Inflation and the Taxation of Savings (1)

Most countries tax **nominal** interest income.

Key definitions:

- **Nominal interest rate**: interest earned in currency terms.
- **Real interest rate**: nominal interest minus inflation.

Taxing nominal interest rates means:

When inflation is high, the real return to saving can be very low or even negative.

Taxation amplifies this distortion.



Inflation and the Taxation of Savings (2)

- The relationship between real and nominal interest rates is:

$$\text{Real interest rate } (r) = \frac{1 + \text{Nominal interest rate } (i)}{1 + \text{Inflation rate } (\pi)} - 1$$

- Inflation increases the nominal but not the real interest rate, but taxes are levied on nominal interest rates.
- Inflation reduces the real after-tax return on savings.



Inflation Exacerbates Capital Taxation (3)

Table 22-1

Case	Inflation	Tax Rate on Interest	Savings	Nominal Interest Rate	Interest Earnings	After-Tax Resources	Price of Skittles	Bags of Skittles
No inflation	0%	0%	100	10%	\$10	\$110	\$1.00	110
	0%	50%	100	10%	\$10	\$105	\$1.00	105
Inflation	10%	0%	100	10%	\$10	\$110	\$1.10	100
	10%	50%	100	10%	\$10	\$105	\$1.10	95.5
Constant real rate	10%	0%	100	21%	\$21	\$121	\$1.10	110
	10%	50%	100	21%	\$21	\$110.5	\$1.10	100.5



Save \$100, 10% real interest rate, 50% tax on interest.

Alternative Models of Saving

The traditional intertemporal model assumes:

- Full rationality
- Perfect foresight
- No borrowing constraints
- Single discount rate

These assumptions may not describe real households.
Gruber presents alternative models to explain observed saving patterns and policy responses.



Liquidity Constraints and Precautionary Saving

Some individuals face binding **borrowing constraints**.

If households cannot borrow against future income:

- Consumption becomes sensitive to short-term income shocks.
- Saving behaves like a buffer against uncertainty.
- Interest-rate changes affect savings less than predicted by basic theory.

Precautionary motives imply:

- High saving among risk-averse or uncertain households.
- Small responsiveness to after-tax interest rates.



Buffer-Stock Behavior

Borrowing constraints and income risk lead to **buffer-stock saving**:

- Households target a minimum level of assets to self-insure.
- Saving adjusts mainly to maintain this buffer.
- Changes in after-tax interest rates have **limited influence**.

This offers an alternative explanation for weak saving responses to capital taxation.



Self-Control and Behavioral Models

Individuals may have **present-biased preferences**:

- Value today's consumption disproportionately.
- Struggle to follow long-term saving plans.
- Choose too little saving relative to long-run goals.

Behavioral models predict:

- Low responsiveness to interest-rate changes.
- Significant effects of commitment devices and defaults.



Commitment Devices and Defaults

Because of self-control problems, many households rely on commitment mechanisms:

- Automatic payroll deductions
- Early withdrawal penalties (401(k), IRA)
- Default contribution rates
- “Save More Tomorrow” automatic escalation

Evidence shows:

- Defaults have large effects on participation and contributions.
- These policies matter **more** for saving than after-tax returns.

This helps explain why tax incentives often shift where people save, but not how much.



Tax-Advantaged Retirement Accounts

Governments encourage private saving by offering tax-preferred accounts.

In the U.S.:

- **401(k) plans** = Tax-preferred retirement savings vehicles offered by employers, to which employers will often match employees' contributions.
- **Individual Retirement Accounts (IRA)** = A tax-favored retirement savings vehicle primarily for low- and middle-income taxpayers who make pre-tax contributions and are then taxed on future withdrawals.

Key economic question:

Do tax subsidies increase total saving, or do they simply shift saving across accounts?



Employer-Sponsored Retirement Plans

Two main types of retirement plans in the U.S.:

- **Defined Benefit (DB):** benefits based on earnings and years of service.
- **Defined Contribution (DC):** benefits depend on accumulated contributions and investment returns.

401(k) plans are the most common DC plan.

They allow tax-deferred contributions up to annual limits.



Traditional IRA and 401(k): Tax Treatment

- Contributions are excluded from taxable income.
- Funds accumulate untaxed.
- Withdrawals are taxed as income in retirement.

Tax benefit:

- Raises the return to saving for eligible individuals.
- Creates an incentive to contribute *up to* the contribution limit.

This tax exclusion creates a **kinked budget constraint**.



Budget Constraint with an IRA (1)

With an IRA or 401(k), individuals face two saving options:

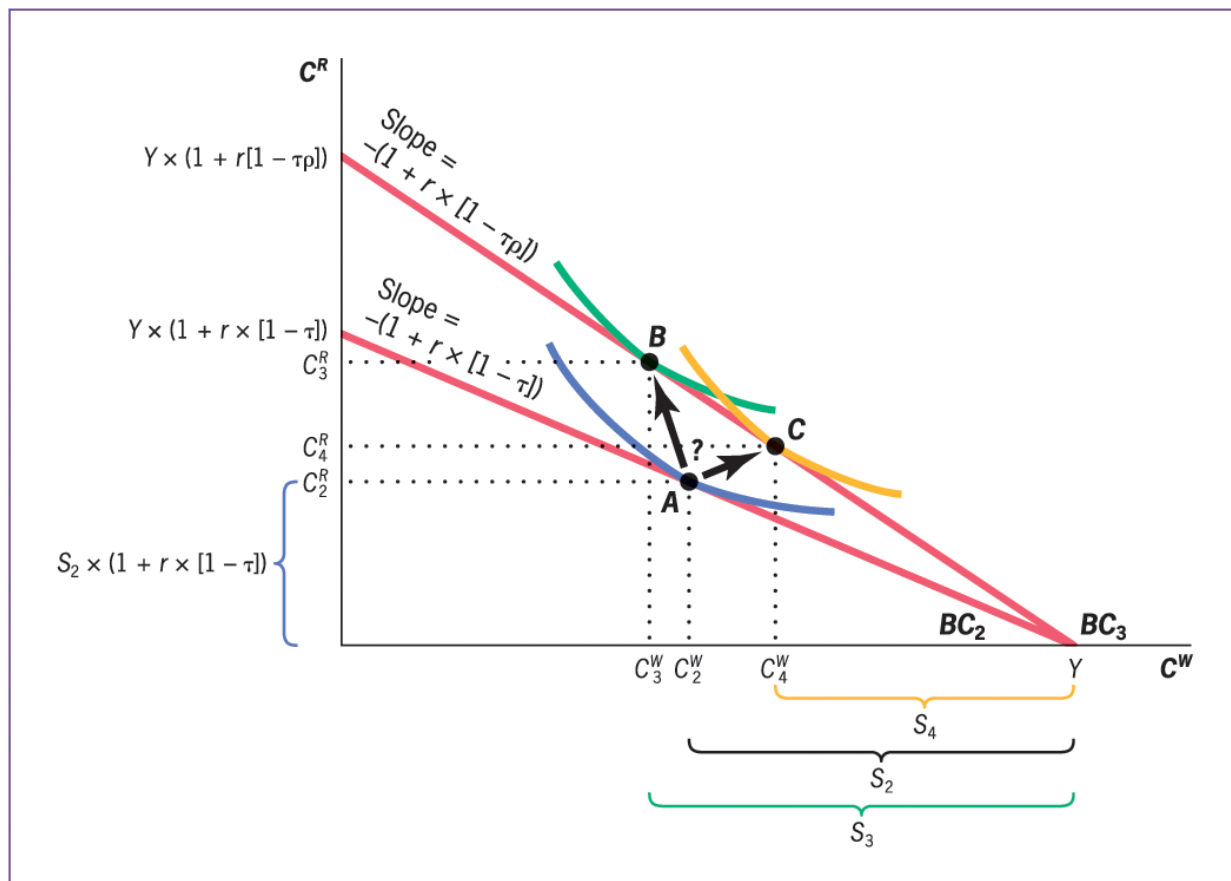
- Save normally: pay tax on interest each year.
- Save in the IRA: no tax on contributions or accumulation; tax applies at withdrawal.

For low levels of saving, individuals can place all saving inside the tax-preferred account, enjoying higher after-tax returns.



Budget Constraint with an IRA (2)

Figure 22-3



© Worth Publishers

Individuals initially face a budget constraint BC_2 with a slope $-(1 + r \times [1 - \tau])$. When retirement savings are tax-subsidized, the budget constraint moves to BC_3 , with a higher slope $-(1 + r \times [1 - \tau \times \rho])$. This leads to a substitution effect toward more savings and an income effect toward less savings.

If the substitution effect is larger, then first-period consumption will fall from CW_2 to CW_3 , and savings will rise from S_2 to S_3 . If the income effect is larger, then first-period consumption will rise from CW_2 to CW_4 , and savings will fall from S_2 to S_4 .



Budget Constraint with an IRA (3)

At the contribution limit:

- The budget line becomes **kinked**.
- Additional saving above the limit must occur in a normal taxable account.

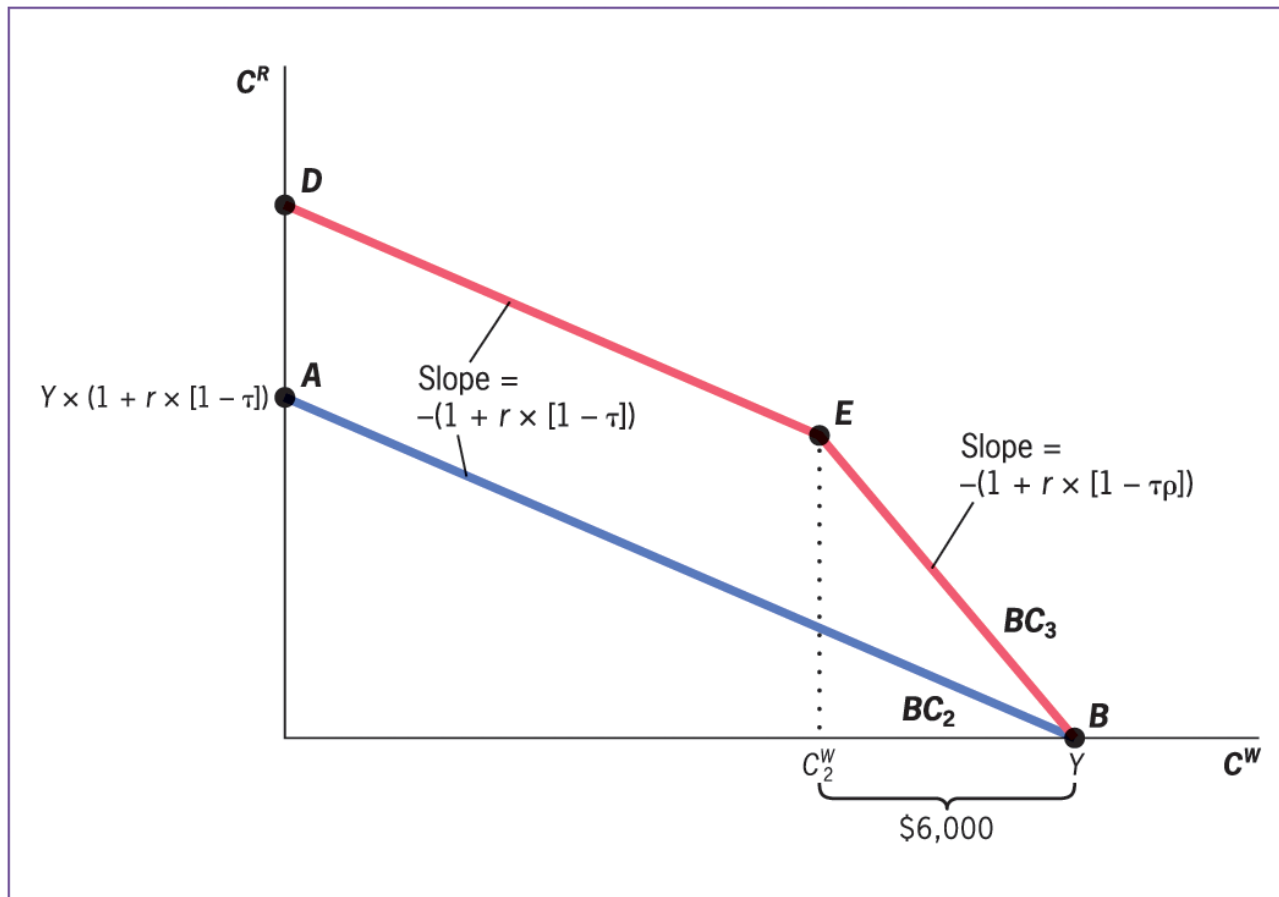
Implication:

The **marginal** return to saving is higher inside the IRA, but **inframarginal** saving simply receives a windfall without increasing total saving.



Budget Constraint with an IRA (4)

Figure 22-4



The availability of IRAs raises the return to savings less than \$6,000 from $-(1 + r \times [1 - \tau])$ to $-(1 + r \times [1 - \tau \times r])$, where r is the net tax preference from using an IRA. Once savings is above \$6,000, the IRA simply increases period-two income, and the return to each dollar of savings returns to $-(1 + r \times [1 - \tau])$.



Marginal vs Inframarginal Responses

IRA tax advantages apply only **up to the contribution limit**.

Two types of savers:

- **Low savers:** save less than the contribution limit → all saving is marginally subsidized.
- **High savers:** save above the limit → only marginal saving faces the higher after-tax return.

This matters for predicting the effect of tax subsidies.



Private vs National Saving

Tax subsidies may increase **private saving**, but this may not translate into higher **national saving** if:

- Government revenue falls, increasing deficits.
- Individuals shift assets across accounts.

Key empirical question:

Do IRAs and 401(k)s raise total saving or simply re-label existing saving?



Roth IRA: Alternative Tax Treatment

Roth IRAs reverse the tax timing:

- Contributions are made **after tax**.
- Accumulation and withdrawals are tax-free.

Economic implications:

Equivalent to a traditional IRA for many savers if tax rates remain constant.

Main difference: timing of revenue and political incentives.

Roth IRAs illustrate how governments use timing of taxation to shape saving incentives.



Empirical Evidence: Key Questions

Do tax-preferred savings accounts actually increase total saving?
Or do they simply re-label funds that individuals would have saved anyway?

Two central empirical questions:

- **Elasticity question:** Do individuals increase total saving when after-tax returns rise?
- **Substitution question:** Do individuals shift savings between accounts with different tax treatments?

Modern evidence uses quasi-experimental variation to answer these questions.



Chetty et al. (2014): Denmark Retirement Reform

A major Danish reform cut the tax subsidy for retirement contributions for high-income workers.

Key features:

- Only affected **top-bracket taxpayers**.
- Allowed clean comparison with lower-income workers (control group).
- Large administrative dataset covering the entire population.

This is one of the most important empirical papers on saving behavior.



Elimination of Subsidy Strongly Reduced Contributions

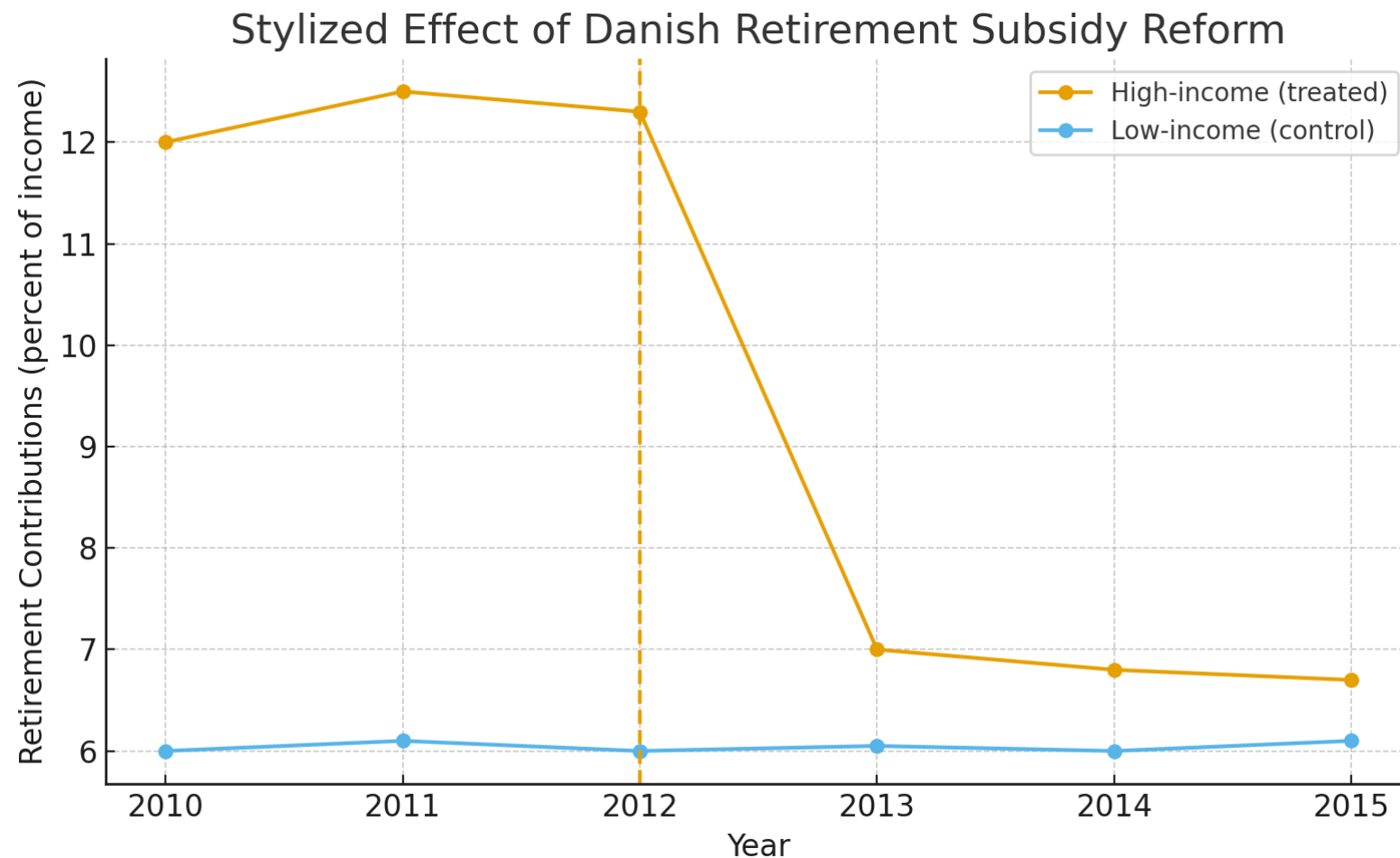
After the reform:

- Contributions among high-income workers **fell sharply**.
- Contributions among unaffected lower-income workers **did not change**.
- The change was almost immediate and very large in magnitude.

Interpretation:

Tax incentives do affect *where* people save.





Source: Chetty, Raj, John N. Friedman, Søren Leth-Petersen, Torben Heien Nielsen, and Tore Olsen (2014).
“Active vs. Passive Decisions and Crowd-Out in Retirement Savings Accounts: Evidence from Denmark.”
Quarterly Journal of Economics, 129(3): 1141–1219.

No Increase in Total Savings

Despite large changes in retirement account contributions:

- **Total private savings did not increase.**
- Individuals compensated by increasing saving in other accounts.
- The fall in retirement saving was almost perfectly offset by increases in taxable saving.

Implication:

Tax subsidies mostly cause **reshuffling**, not new saving.



Mandated Saving Has Large Effects

In contrast to subsidies:

- Danish **mandatory** employer retirement contributions increased total saving one-for-one.
- Workers who were required to save more did not reduce saving elsewhere.

Mandates are much more effective than subsidies at raising total saving.



Active vs. Passive Savers

Chetty et al. classify individuals into two groups:

Active savers

- Adjust savings across accounts.
- Respond strongly to price incentives.
- Substitution dominates.

Passive savers

- Contribute set amounts and rarely change behavior.
- Do not offset mandatory contributions.
- Default rules matter greatly.

Result:

Most individuals are **passive savers** → this explains the large effect of mandates and defaults.



Policy Implications of Active vs Passive

Tax incentives mainly influence **active savers**, who tend to be higher-income.

Defaults, automatic enrollment, and mandates influence **passive savers**, who are the majority.

Therefore, subsidies are:

- Less effective at raising total savings
- More regressive
- Less cost-effective than automatic mechanisms

This modern view shapes policy debates in Europe and the U.S.



International Stylized Facts

Key cross-country insights on capital taxation and saving:

- Capital income tax rates vary widely across OECD countries.
- Countries with generous mandatory retirement systems show **low sensitivity** of saving to tax incentives.
- Capital shares and wealth concentration have risen globally.
- Despite different tax structures, **household saving rates are relatively stable** across countries.

This evidence reinforces the idea that deep behavioral and institutional factors dominate over tax incentives.

Source: Saez (Berkeley Public Economics Lectures); Saez & Zucman (2019).



Modern View of Tax Incentives for Saving

Summary of the empirical consensus:

- Substitution across accounts is large.
- Effects on **total saving** are small or zero.
- Mandates and default rules have strong effects.
- Behavioral factors matter more than after-tax returns.
- High-income individuals respond most to tax incentives.
- International differences reflect mandatory systems and household portfolios.

Source: Saez (Berkeley Public Economics Lectures); Saez & Zucman (2019).

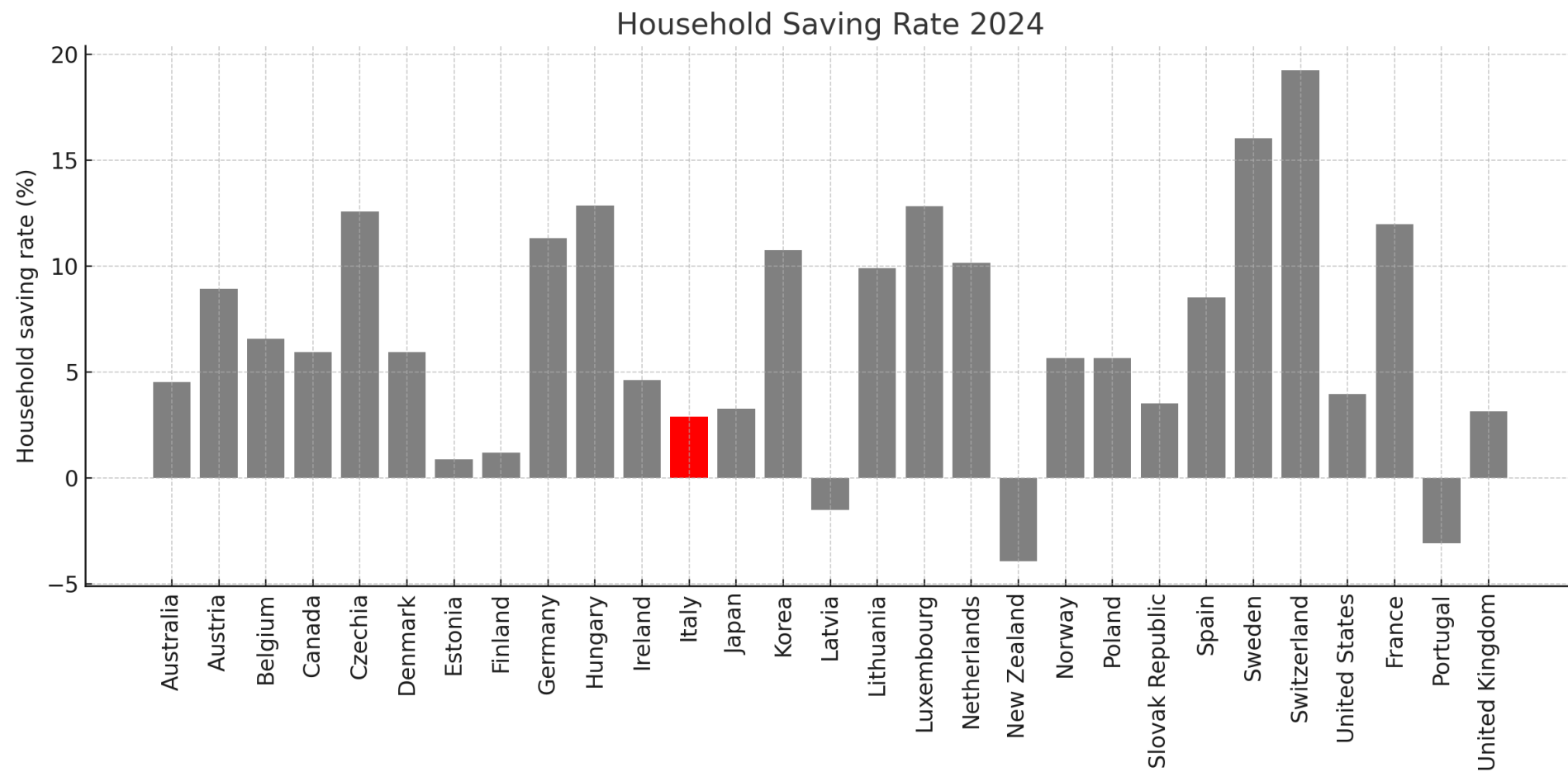


Italy: Household Saving Rate

- Italy's household saving rate has declined steadily since the 1990s.
- Rates remain lower than in Germany and France, but similar to Spain.
- Demographic aging and slow income growth contribute to low saving.
- Implication: responsiveness to tax incentives is likely limited.

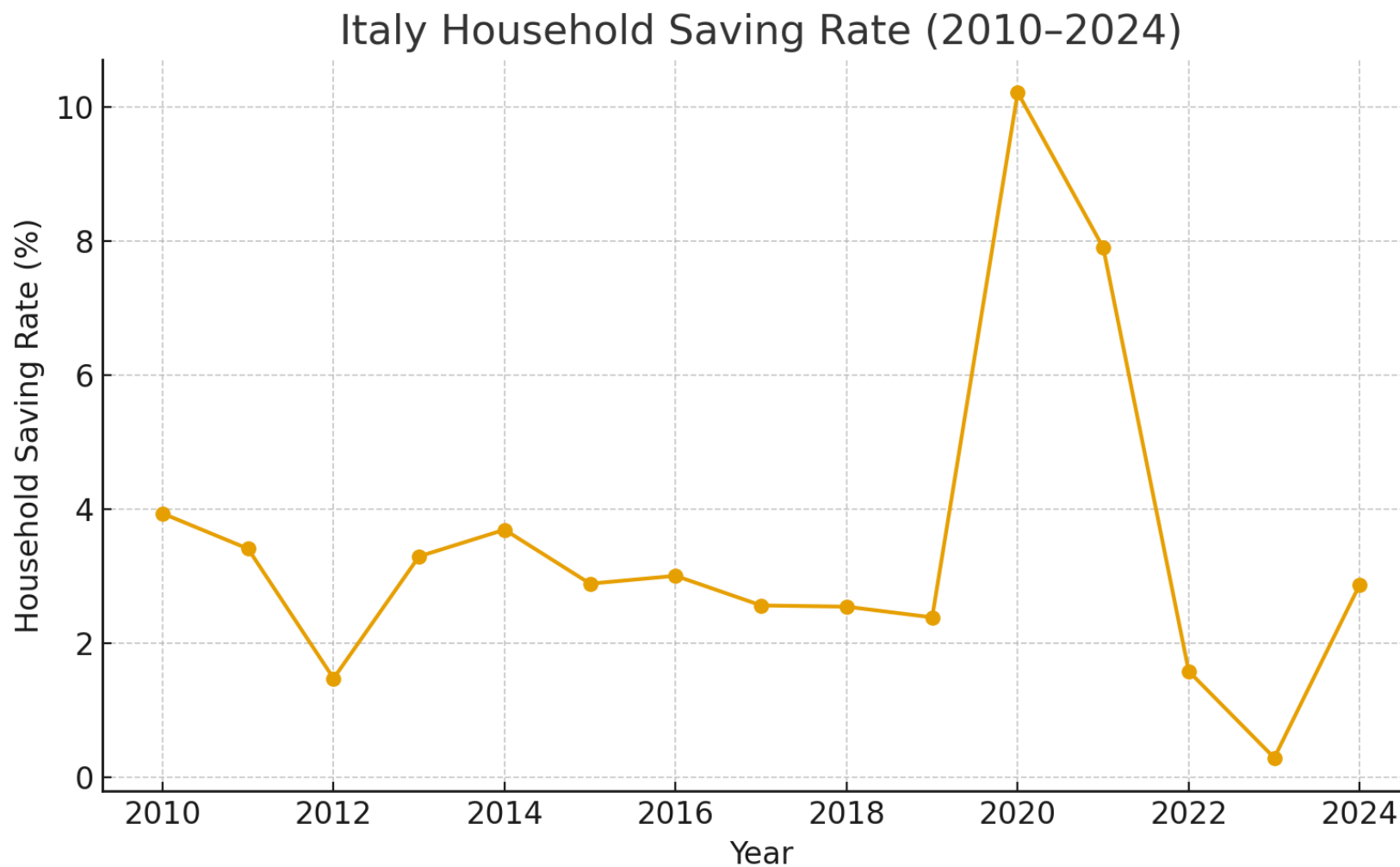
Source: OECD Household Saving Rate database.



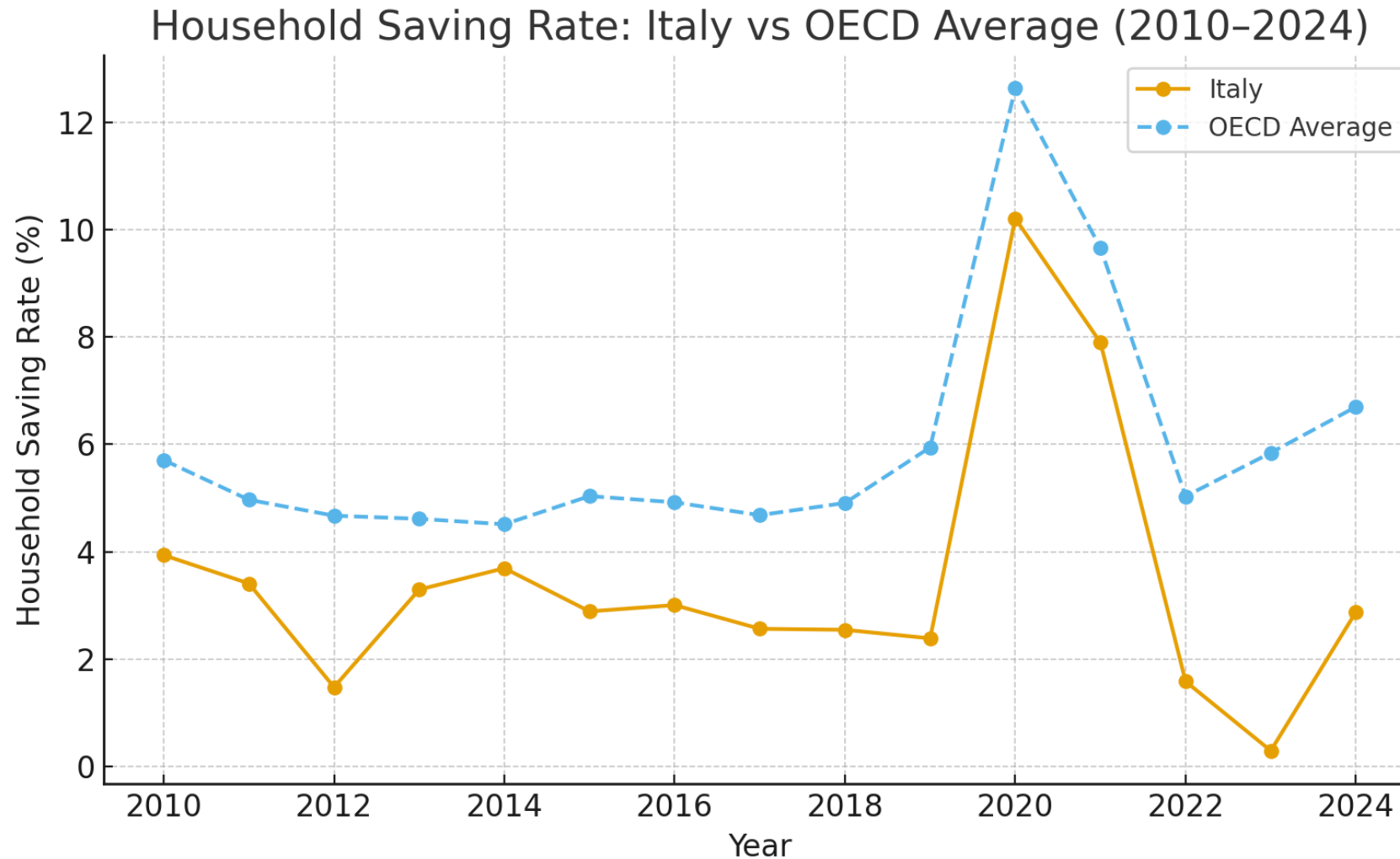


Source: OECD Economic Outlook Database (EO115), *Annex Table: Household Saving Rates*.





Source: OECD Economic Outlook Database (EO115), *Annex Table: Household Saving Rates*.



Source: OECD Economic Outlook Database (EO115), *Annex Table: Household Saving Rates*.

Exercises



MC 1

Who is most responsive to tax incentives for saving?

- a) Passive savers
- b) Low-income liquidity-constrained households
- c) High-income active savers
- d) Retirees



MC 2

According to the intertemporal choice model, why does a tax on interest income have an *ambiguous* effect on saving?

- a) Because saving depends only on preferences, not on interest rates
- b) Because the substitution effect lowers saving while the income effect raises saving
- c) Because both substitution and income effects always reduce saving
- d) Because borrowing constraints eliminate both effects



MC 3

Why do tax-preferred retirement accounts (such as IRAs or 401(k)s) often fail to increase *total* saving?

- a) Because contribution limits are too low
- b) Because only low-income households use them
- c) Because individuals often shift existing savings into these accounts rather than increasing total saving
- d) Because they reduce the after-tax return to saving



MC 4

Evidence from Chetty et al. (2014) shows that eliminating a subsidy for high-income savers in Denmark:

- a) Greatly reduce retirement contributions and reduced total savings
- b) Had no effect on contributions
- c) Reduced contributions but total savings remained nearly unchanged
- d) Increased total savings



MC 3

Which policy is most likely to generate a one-for-one increase in total saving?

- a) A reduction in tax rates on interest income
- b) Expanding contribution limits for IRAs/401(k)s
- c) Introducing a mandate for employer retirement contributions
- d) Offering an informational campaign about financial literacy



True/False (please provide a justification)

- 1) A reduction in the tax rate on interest income necessarily increases national saving.
- 2) Passive savers respond strongly to changes in after-tax interest rates.
- 3) Taxing nominal interest income can lead to negative real after-tax returns when inflation is high.



Exercises

EXERCISE 1

Suppose Italy reduces the tax rate on interest income from 26% to 20%.

- Predict the effect on saving for:
 - ☐ liquidity-constrained households
 - ☐ high-income active savers
 - ☐ passive savers

b) Would national saving necessarily rise?

c) Denmark-style mandates increased saving one-for-one. Would a similar mandate be effective in Italy?

d) How might Italy's portfolio structure affect the result?



Exercises

EXERCISE 2

In the intertemporal choice model, individuals choose consumption today (C_1) and consumption in the future (C_2). Suppose a tax on interest income is introduced, lowering the after-tax interest rate. Suppose the interest rate is 10% and the tax rate is 20%.

- a) How much future consumption does giving up 1 unit of consumption today generate before and after the tax?
- b) Based on your answer in (a), explain how the tax affects the incentive to save.
- c) Which effect (substitution or income) is generated by this change? Briefly explain.

